

A new approach to estimate the spatial distribution of solar radiation using topographic factor and sunshine duration in South Korea



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ABSTRACT

Solar radiation is one of the important renewable resources, currently scientists are taking their interest in. Accurate solar radiation data is not only required for solar-power management but also is a vital input parameter in different biogeochemical and atmospheric models. But there are inadequate number of stations measuring solar radiation in comparison to stations dedicated for sunshine duration, temperature, humidity etc. Therefore, to overcome this problem, an empirical model is developed to estimate solar radiation from sunshine duration data over South Korea. As more than 50% of the area in Korean peninsula have a complex terrain, a topographical factor is applied to modeled data. Thereafter a map presenting monthly mean variation in incoming solar insolation is constructed using ordinary kriging method. The influence of topographical features like slope and aspect is found to be higher in winter than summer. Solar radiation is highest in May and lowest in December over Korea. Spatial variation of incoming radiation is mainly influenced by topographical and atmospheric features whereas latitudinal gradient is almost insignificant.

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1. Introduction

The growing threat of global warming and exhaustion of current fossil fuel resources have motivated the global scientific community to search and develop other non-conventional energy sources like solar energy. Although in each second the amount of solar energy reaching the earth surface is 10,000 times higher than the current daily energy requirements, it constitute only 0.1% of total world consumption, mainly due to technical inefficiency and large developmental cost [1]. However, recently the importance of solar energy has again been highlighted, due to increasing pressure from Intergovernmental Panel on Climate Change (IPCC) to reduce the global carbon dioxide emission. Besides a number of human economical activities like building design, land management, agriculture and forestry ultimately depend on incoming solar radiation. It is also the major driving force behind earth's different biological and atmospheric phenomenon. Therefore, besides assisting in site selection, planning and simulation of solar energy system, accurate insolation data is also a key component in undertaking many renewable, hydrological and agricultural studies. Moreover a precise map of available solar energy is extremely

important in decision making support system not only for selection of right land for related facilities but also for maximizing the efficiency of solar energy transfer. A simple interpolation method can be used to generate an insolation map from ground based measurements [2–6]. But in most areas of the world, solar radiation measuring stations are not easily available due to financial, technical or institutional limitations [7]. The insufficient number affects the spatial and temporal resolution of such image which in turn leads to incorrect system sizing and unnecessary use of conventional energy. The number of weather stations recording daily radiation is minor compared to the number which records other meteorological parameters. The global ratio between stations measuring radiation and those measuring temperature may be as low as 1:500 [8]. In South Korea there are approximately 460 Automatic Weather Stations (AWS) and 96 weather stations for measuring temperature and precipitation, among which solar radiation is measured only in 22 weather stations. Hence, an empirical model relating solar radiation with other atmospheric parameters can be helpful in generating more set of points with available data which successively can enhance the resolution of insolation map. There are many studies to estimate solar radiation based on empirical relationships between the observed solar radiation and the meteorological elements. The empirical models are mainly based on temperature [8–13] cloudiness [14–16] and sunshine duration [5,17–23] or combination of them [24]. The sunshine based

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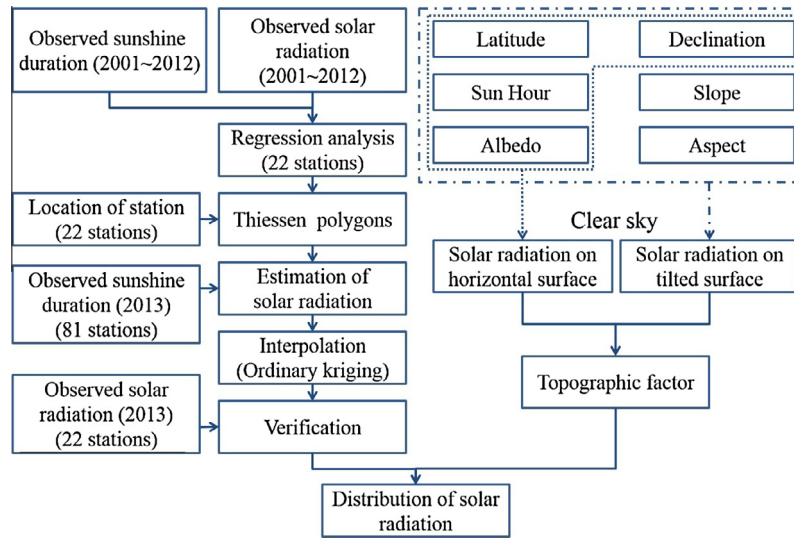


Fig. 1. Flow chart of global solar radiation distribution model.

methods are generally more accurate and therefore are commonly used for estimating solar radiation by many researchers [25–27].

However, at the landscape scale, variability in elevation, surface orientation (slope and aspect) and shadows cast by topographic features are the major factors modifying the distribution of radiation. Different scientists developed a variety of methodologies to integrate topographical features in estimation of ground radiation. Que et al. [28] studied a distribution model for calculating extraterrestrial solar radiation over rugged terrain based on Digital Elevation Model (DEM). Kumar et al. [29] described the potential solar radiation on any slope and aspect and at many latitudes. Pons and Miquel [30] has developed and evaluated a methodology for obtaining monthly radiation maps based on DEM, field data and GIS tools. Ruiz-arias et al. [31] analyzed the solar radiation model using DEM and topographic characteristics. Having a mountainous terrain, in South Korea the topographic factors can create a strong localized gradient in available solar energy. The slope average in the South Korea is 14.3 degrees, and 45.2% is lowland area with less than or equal to 15 degrees and 30.7% is more than 20 degrees [32]. Thus, for more accuracy in estimating solar radiation it is necessary to consider the topographic conditions such as slope and aspect.

The purpose of the present study is to estimate the regional and temporal distribution of solar radiation over South Korea based on the spatial interpolation of ground data. For this reason, empirical relationships have been developed to derive the solar radiation at horizontal surface using sunshine duration data. The changes in the solar radiation for surface slope and aspect has also been included to investigate the influence of topographic features.

2. Methods and materials

In this section a schematic description of the steps followed to estimate the solar radiation is reported. At first, 22 stations collecting both solar radiation and sunshine data from 2001 to 2012 is selected for regression analysis. After that, the Thiessen polygons are created using these 22 stations. It divided the other 59 stations measuring only the sunshine hours into 22 groups. Each group having only one radiation observing station has the same regression parameters. Thereafter, insolation data over all these 81 stations are taken into account for spatial interpolation. But it only derived the spatial distribution of incoming radiation at horizontal surface. The topographic factor is modeled using the effect of surface slope and aspect on clear sky solar radiation and it is then

applied on the interpolated spatial distribution to construct accurate solar radiation data over complex terrain. A simple flowchart summarizing our approach towards the estimation of a precise solar radiation map over South Korea is depicted in Fig. 1.

2.1. Study area

The present study is conducted over the Republic of Korea which occupies the southern portion of the Korean peninsula. It lies between latitudes 33°N and 39°N, and longitudes 124°E and 130°E covering a total area of 100,032 km². The South Korea's terrain is mostly mountainous, most of which is not arable. Lowland, located primarily in the west and southeast, make up only 30% of the total land area. Slope average in the South Korea is 14.3 degree, and lowland area with less than 5 degree of slope is only 28.2%. Otherwise, 58.8% of total area is more than 10 degree of slope, 45.2% is more than 15 degree and 30.7% is more than 20 degree [32]. Hence, for accurate measurement of solar radiation consideration topographic conditions such as slope and aspect is extremely essential.

2.2. Regression model

The history of solar energy estimation algorithms using meteorological parameters dates back to 100 years. One of the first work published by Ångström [16] expressed the correlation between global solar radiation and sunshine duration in the sky. Sunshine-based model using Ångström-Prescott equation, when tested for 59 localities covering Europe, Asia, Africa, and North America, showed an average Mean Bias Error (MBE) of only 8.8% [33]. The equation established an empirical relationship between clearness index (K_t) and relative sunshine duration as shown in Eq. (1) [25,26].

$$K_t = \frac{H}{H_0} = a + b \left(\frac{n}{N} \right) \quad (1)$$

Here H is solar radiation on horizontal surface at ground (MJ m⁻² day⁻¹), H_0 is extraterrestrial radiation on a horizontal surface (MJ m⁻² day⁻¹), n is hours of sunshine (hour day⁻¹), N is hours of maximum possible sunshine (hour day⁻¹) and a , b are the regression constants to be determined.

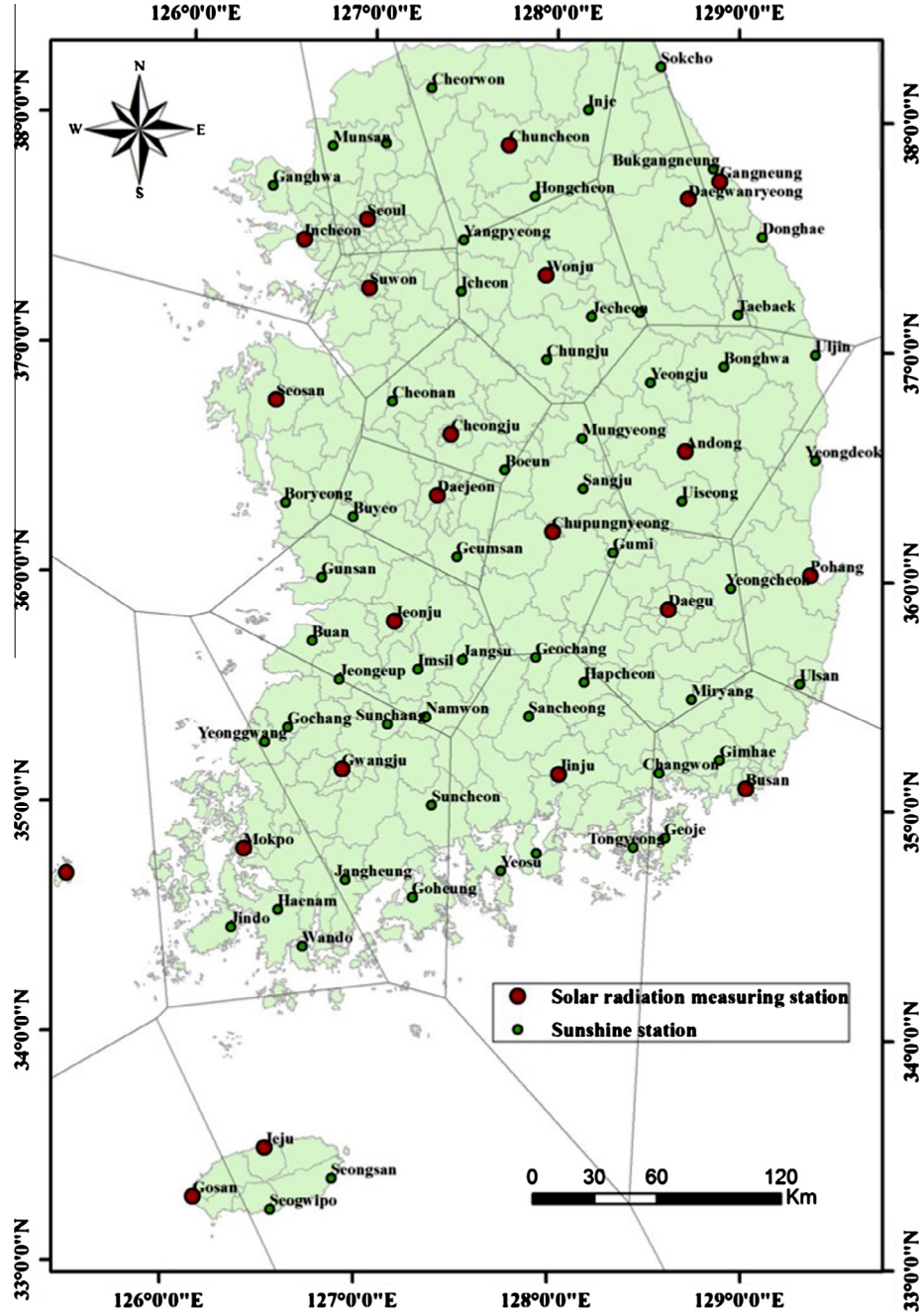


Fig. 2. Thiessen polygons centering solar radiation stations.

Assuming that in a day the Sun–Earth distance is constant, the density of solar flux energy incident on a horizontal surface is computed from the following equation (Eq. (2)) [34].

$$H_0 = \frac{24 \times 3600 G_{sc}}{\pi} \left(1 + 0.033 \cos \frac{360 \times DOY}{356} \right) \times (\cos L \cos \delta \sin \omega + \omega \sin L \sin \delta) \quad (2)$$

Here G_{sc} is solar constant (1373 W m^{-2}), DOY is the day of year, L is latitude of the station (degrees), δ is declination angle (degrees) and ω is sunrise hour angle (degrees).

The declination, δ , for any day of the year can be calculated approximately by the following equation,

$$\delta = 23.45 \sin \left(360 \frac{284 + DOY}{365} \right) \quad (3)$$

The sunset hour angle ω is given as,

$$\omega = \cos^{-1} (-\tan L \tan \delta) \quad (4)$$

Furthermore, the hours of possible sunshine, N is expressed by,

$$N = \left(\frac{2}{15} \right) \omega \quad (5)$$

In the present work, H_0 and N were computed for each weather station by using Eq. (2) and (5). The regression coefficients a and b in Eq. (1) have been calculated from observed values of H/H_0 and n/N from 2001 to 2012.

2.3. Weather stations and Thiessen polygons

There are a total of 97 weather stations including 10 radar, 4 upper-air observation stations, and approximately 460 AWS

Table 1
Weather stations of Thiessen group.

Group	Radiation stations	Sunshine stations
1	Incheon	Ganghwa, Baengnyeongdo
2	Seoul	Dongducheon, Munsan
3	Chuncheon	Yangpyeong, Inje, Cheorwon, Hongcheon
4	Daegwanryeong	Sokcho, Taebaek
5	Gangneung	Donghae, Bukgangneung
6	Seosan	Boryeong
7	Suwon	
8	Wonju	Suwon, Yeongwol, Incheon, Jecheon, Chungju
9	Cheongju	Boeun, Cheonan
10	Andong	Bonghwa, Yeongju, Uljin, Uiseong
11	Pohang	Yeongdeok, Ulsan
12	Daejeon	Geumsan, Buyeo
13	Jeonju	Gunsan, Namwon, Buan, Imsil, Jangsu, Jeongeup
14	Chupungnyeong	Gumi, Mungyeong, Sangju
15	Daegu	Miryang, Yeongcheon
16	Heuksando	
17	Mokpo	Wando, Jindo, Haenam
18	Gwangju	Gochang, Goheung, Sunchang, Suncheon, Yeonggwang, Yeonggwang
19	Gosan	
20	Jeju	Seogwipo, Seongsan
21	Jinju	Geochang, Namhae, Sancheong, Yeosu, Tongyeong, Tongyeong
22	Busan	Geoje, Gimhae, Changwon

operated by the Korean Meteorological Administration (KMA) in South Korea [35]. In this study, 22 out of a total 97 weather stations were selected as they hold long-term records (1965 to 2013) for solar radiation, and the data reflecting sunshine duration which was collected from 81 stations from 2001 up to 2013. Thiessen method was used for estimating solar radiation from 59 weather stations measuring only sunshine duration. The Thiessen method calculates average solar radiation across the region by multiplying each station's value by the area of its polygon and dividing by the total area. The Thiessen map suggests that model parameters at all locations within each polygon belongs to a particular weather station within that polygon and therefore, has the same value (Fig. 2). Table 1 shows that 59 weather stations without solar radiation data and corresponding radiation measuring stations.

2.4. Topographic factor

The amounts of cloud, DOY, latitude, sun position, topography are some of the important factors which can influence the solar radiation level [36]. The topographic factor or TF is the ratio of global solar radiation in a tilted surface (H_t) and at horizontal surface (H). It is modeled using surface slope, aspect, latitude and basic solar parameters with the help of following equations. Total solar irradiance on a tilted surface (H_t) with slope β is calculated as a sum of three terms:

$$H_t = H_{Bt} + H_{Dt} + H_{Gt} \quad (6)$$

The H_t consists of direct (H_{Bt}), diffuses (H_{Dt}) and reflected (H_{Gt}) solar radiation on a tilted surface (Eq. (6)). The H_{Bt} reaches the surface of Earth from the solar beam unaffected by atmospheric gases and particles. On the other hand, H_{Dt} comprises the part of the solar radiation scattered by small gas molecules, dust particles and aerosols. The H_{Gt} is mainly reflected from terrain and is therefore more important in mountainous areas.

Solar radiation on a tilted surface mainly depends on slope, aspect, solar altitude and azimuth. The solar altitude and azimuth exhibit both day to day variations as well as variation with the time of day and therefore demands rigorous calculations. Whereas, the slope and aspect of the terrain are fixed and simple to estimate. Therefore topographic factors of direct solar radiation

were computed using the ratio of clear sky solar radiation on a horizontal and tilted surface excluding the effect of cloud (Eq. (7)). In case of a plain land, the solar incident angle (θ) will be as same as solar zenith angle (ϕ) and consequently, the H_{Bt} will be equal to H_B .

$$TF_B = \frac{H_{Bt}}{H_B} = \frac{\cos \theta}{\cos \phi} \quad (7)$$

$$\begin{aligned} \cos \theta = & (\sin L \sin \delta \cos \beta - \cos L \sin \delta \sin \beta \cos Z_s + \cos L \\ & \times \cos \delta \cos h \cos \beta + \sin L \cos \delta \cos h \sin \beta \cos Z_s \\ & + \cos \delta \sin h \sin \beta \sin Z_s) \end{aligned} \quad (8)$$

$$\cos \phi = (\sin L \sin \delta + \cos L \cos \delta \cos h) \quad (9)$$

Here TF_B is topographic factor of direct solar radiation, H_{Bt} is direct solar radiation on tilted surface, H_B is direct solar radiation on horizontal surface, θ is solar incidence angle, ϕ is solar zenith angle, β is surface tilt angle from the horizon, Z_s is surface azimuth angle, also known as topographical aspect and h is hour angle.

An isotropic sky model originally developed by Hottel and Woertz [37] and modified by Liu and Jordan [38] is used to estimate the H_t . Diffuse solar radiation on horizontal surface H_D and on tilted surface H_{Dt} were defined by Eqs. (10) and (11) respectively.

$$H_D = 2 \int_0^{\frac{\pi}{2}} H_R \cos \phi d\phi = 2H_R \quad (10)$$

$$H_{Dt} = H_D \left(\frac{1 + \cos \beta}{2} \right) \quad (11)$$

Here, H_R is diffuse radiation received from sky dome.

Reflected solar radiation on horizontal surface H_G and on tilted surface H_{Gt} can be expressed by the following equations,

$$H_G = \rho_G (H_B + H_D) = 2 \int_0^{\frac{\pi}{2}} H_r \cos \phi d\phi = 2H_r \quad (12)$$

$$H_{Gt} = \int_{\frac{\pi}{2}-\beta}^{\frac{\pi}{2}} H_r \cos \phi d\phi = \rho_G (H_B + H_D) \left(\frac{1 - \cos \beta}{2} \right) \quad (13)$$

Here ρ_G is ground albedo, G_r is the isotropic reflected radiation.

The global solar radiation on tilted surface can be written as Eq. (14). The global solar radiation on horizontal surface (H) is the summation of direct and diffuse part of the solar radiation as expressed in Eq. (15). Therefore, the topographic factor of global solar radiation, TF can be expressed by Eq. (16).

$$H_t = TF_B H_B + H_D \left(\frac{1 + \cos \beta}{2} \right) + \rho_G (H_B + H_D) \left(\frac{1 - \cos \beta}{2} \right) \quad (14)$$

Table 2
Seasonal variation of bright sunshine hours over South Korea.

	N^a	Average sunshine hours	SD ^b	Kurtosis	CV
Jan	9.84	5.56	3.35	-1.20	0.60
Feb	10.71	6.08	3.78	-1.28	0.62
Mar	11.76	7.76	3.12	-0.18	0.40
Apr	12.93	7.19	4.00	-0.92	0.56
May	13.92	7.78	4.47	-1.13	0.57
Jun	14.42	5.98	4.29	-1.38	0.72
Jul	14.19	4.65	4.02	-1.14	0.86
Aug	13.34	7.65	3.73	-0.72	0.49
Sep	12.22	6.00	4.16	-1.48	0.69
Oct	11.05	6.86	3.35	-0.56	0.49
Nov	10.07	5.91	3.03	-0.95	0.51
Dec	9.59	5.50	3.06	-1.02	0.56

^a N is maximum possible sunshine hours.

^b SD is standard deviation.

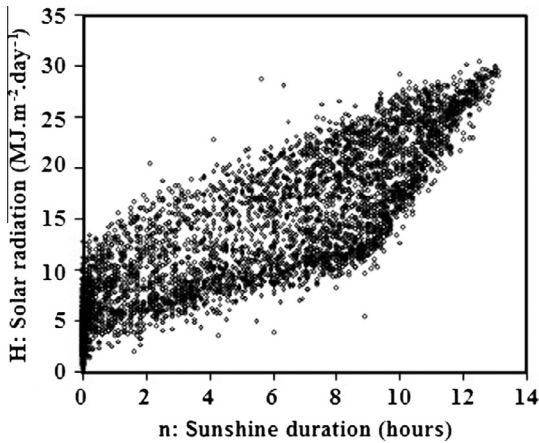


Fig. 3. Variation of solar radiation with sunshine hours over Mokpo.

$$H = H_B + H_D \quad (15)$$

$$TF = \frac{H_t}{H} = \frac{H_B}{H} TF_B + \frac{H_D}{H} \left(\frac{1 + \cos \beta}{2} \right) + \rho_G \left(\frac{1 - \cos \beta}{2} \right) \quad (16)$$

The TF is computed for each 30 m × 30 m ground area (pixel) and then applied to the interpolated solar radiation (at horizontal surface) to estimate the radiation on a tilted surface (Fig. S1).

2.5. Interpolation

Kriging, named after Dr. Danie G. Krige, is a geo-statistical method for interpolation based on Gaussian process. It generates intermediate values from a set of scattered points with observed values. Mathematically it is similar to regression analysis as it give the best linear unbiased estimate at the intermediate points. It differs from the other spatial interpolation methods like Inverse Distance Weighted (IDW) in that it use a semi-variogram model developed from the spatial structure of the dataset. There are many methods of Kriging such as Simple Kriging (SK), Ordinary Kriging (OK), Universal Kriging (UK), Disjunctive Kriging (DK) and

Co-kriging. But the OK method is the most general and widely used among the Kriging methods. In Ordinary Kriging at first an experimental variogram is created from the scattered set of points by plotting the variance with respect to distance between the points. After that a well-defined modeled variogram is computed using the trend in experimental model [39]. The basic guiding equation used in OK method is as follows,

$$H^*(x_0) = \sum_{i=1}^k \lambda_i H(x_i) \quad (17)$$

Here H^* is solar radiation at a definitive point x_0 , $H(x_i)$ and λ_i are solar radiation and the OK weights applied for i^{th} station and k is the total number of measured stations. It is assumed that the OK weights (λ_i) can be express by $\sum_{i=1}^n \lambda_i = 1$. It is only possible when there will be no biased between estimated and observed data. Generally, as shown in Eq. (18), a semi-variogram function $\gamma(h)$ is used to solve the kriging system of equations.

$$\gamma(h) = \frac{1}{2n} \sum_{i=1}^n [H(x_i) - H(x_i + h)]^2 \quad (18)$$

Here $H(x_i)$ and $H(x_i + h)$ are the observed solar radiation at points x_i and $x_i + h$, respectively. But in reality, two measurement sites are seldom found to be separated by h . Therefore, the data is grouped into pairs with similar separating distance (h). Each distance lag contains n number of pairs. In this study, an exponential model has been used to fit the semi-variograms. Particularly, this model parameterizes the semi-variogram in the following Eq. (19).

$$\gamma(h) = C_0 + C_1 \left[1 - \exp\left(\frac{-h}{a}\right) \right] \quad (19)$$

Here C_0 , C_1 , and a are called, respectively, nugget, sill, and range.

2.6. Accuracy and validation

The validation of modeled solar radiation is carried out using observed insolation data for the year of 2013. Three features of frequently used error analysis Mean Absolute Bias Error (MABE), Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) are computed using Eqs. (20)–(22) respectively.

Table 3
Model parameters and error statistics for 22 radiation measuring stations.

Group	Radiation stations	Regression parameters				Error parameters		
		a	b	$a + b$	R^2	MABE ^a	RMSE ^a	MAPE (%)
1	Incheon	0.55	0.17	0.72	0.85	1.28	1.36	9.89
2	Seoul	0.54	0.19	0.73	0.84	2.21	2.39	16.07
3	Chuncheon	0.58	0.20	0.78	0.87	1.07	1.22	8.18
4	Daegwanryeong	0.60	0.19	0.79	0.89	1.33	1.55	8.62
5	Gangneung	0.59	0.19	0.78	0.90	1.32	1.90	11.12
6	Seosan	0.55	0.21	0.76	0.88	2.18	2.31	14.83
7	Suwon	0.54	0.20	0.74	0.89	0.99	1.34	7.20
8	Wonju	0.57	0.21	0.78	0.88	2.23	2.52	14.36
9	Cheongju	0.54	0.20	0.74	0.84	2.08	2.73	13.57
10	Andong	0.50	0.19	0.69	0.83	0.60	0.77	4.21
11	Pohang	0.57	0.19	0.76	0.90	1.25	1.58	8.95
12	Daejeon	0.57	0.20	0.77	0.88	0.46	0.61	2.96
13	Jeonju	0.55	0.20	0.76	0.90	2.23	2.36	15.29
14	Chupungnyeong	0.57	0.18	0.75	0.85	1.26	1.64	9.09
15	Daegu	0.53	0.20	0.73	0.89	0.51	0.71	3.14
16	Heuksando	0.58	0.21	0.80	0.88	0.70	0.78	5.48
17	Mokpo	0.56	0.21	0.77	0.90	1.93	2.03	13.04
18	Gwangju	0.57	0.21	0.78	0.85	2.11	2.25	13.48
19	Gosan	0.62	0.18	0.80	0.97	1.81	2.14	12.27
20	Jeju	0.60	0.18	0.78	0.89	0.38	0.47	2.90
21	Jinju	0.55	0.19	0.74	0.72	0.59	0.74	3.58
22	Busan	0.57	0.19	0.76	0.87	1.37	1.43	8.48

^a MJ m⁻² day⁻¹.

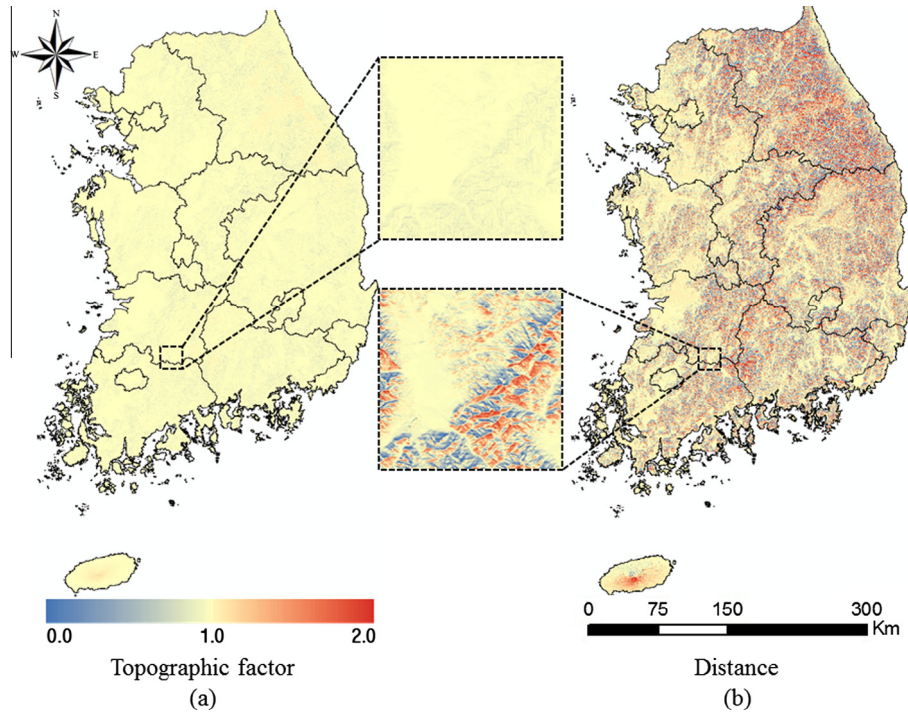


Fig. 4. The topographic factor (a) on June and (b) on December in the South Korea.

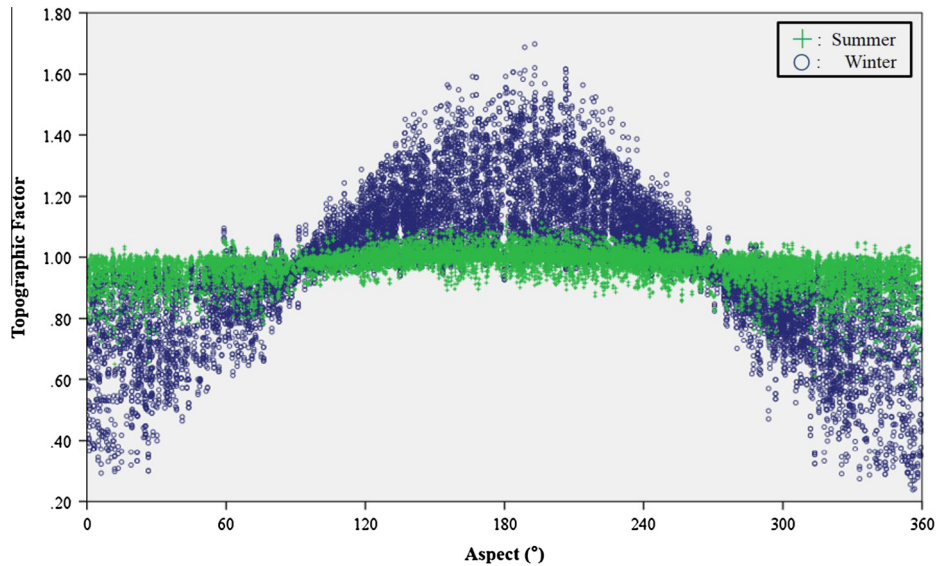


Fig. 5. The variation of topographic factor with aspect for the summer and winter.

$$MABE = \frac{1}{X} \sum_{i=1}^X |H_{ci} - H_{mi}| \quad (20)$$

$$RMSE = \sqrt{\frac{1}{X} \sum_{i=1}^X (H_{ci} - H_{mi})^2} \quad (21)$$

$$MAPE = \frac{1}{X} \sum_{i=1}^X \frac{|H_{ci} - H_{mi}|}{H_{mi}} \times 100 \quad (22)$$

Here, H_c and H_m are respectively the calculated and measured solar radiation and X is the total number of observation.

3. Results and discussion

3.1. Regression equation

South Korean peninsula has a humid temperate climate with maximum precipitation during July. The relative bright sunshine duration which is decrease with increasing cloud cover, is highest during March and lowest during July. As shown in Table 2, a high coefficient of variation (CV) in June and July also indicates a high fluctuation in sunshine (n) data during these months. The data shows a more centralized distribution for the month of March with lowest CV and kurtosis. The bright sunshine hours over South Korea varies from 0 to 14.1 in 2013 with an average 6.41 ± 3.85 .

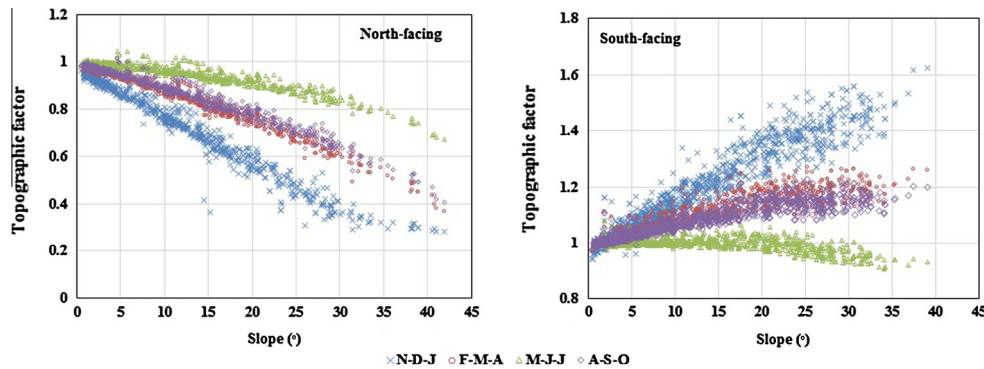


Fig. 6. The variation of monthly topographic factor with slope.

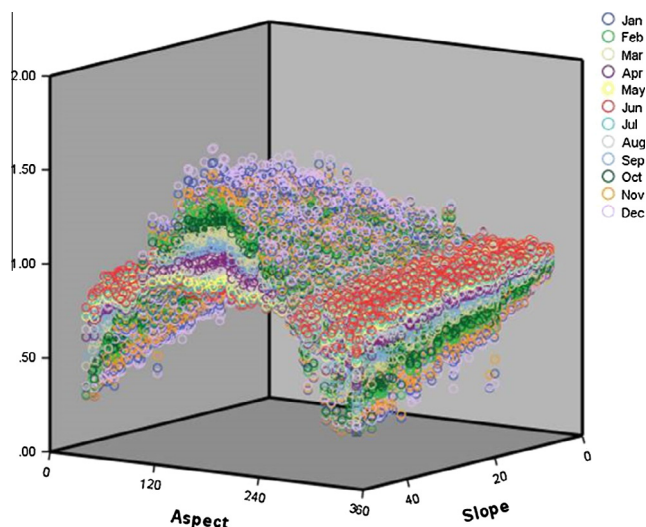


Fig. 7. Influence of slope and aspect on topographic factor.

The variation of observed solar radiation with sunshine hours is given in Fig. 3. For estimating solar radiation the ratio of observed (n) and expected (N) sunshine duration over 22 stations is used as one of the basic model inputs.

The regression and error parameters are given in Table 3. The regression parameters ' a ' ranges from 0.50 to 0.62, while the sum of ' a ' and ' b ' is from 0.69 to 0.80. The steep slope (a) over Gosan and Daegwanryeong stations signifies that the change in solar radiation is much higher over these stations with a unit change in sunshine duration. The slope is generally higher in low latitudinal or high altitudinal stations except some like Gangneung. For validation of the model the expected solar radiation for the year of 2013 is calculated and compared with observed values. The error statistics of the regression equations were in the ranges from 0.38 to $2.23 \text{ MJ m}^{-2} \text{ day}^{-1}$ for MABE, $2.90 \sim 16.07 \text{ MJ m}^{-2} \text{ day}^{-1}$ for MAPE, and $0.47 \sim 2.73 \text{ MJ m}^{-2} \text{ day}^{-1}$ for RMSE. But overall high correlation ($R > 0.8$) between observed and estimated data and low error established that these regression models can be used for solar radiation measurements over other 59 stations having only sunshine data.

3.2. Topographic factor

Two important topographic features, slope and aspect are considered to estimate the influence of topographic condition on solar insolation. Depending on the purpose, the topographic factor (TF) can be calculated for daily, monthly and annual mean solar

radiation. In this study, TF for each month is computed using Eqs. from (6)–(16). It is equal to 1.0 in case of ocean or a partial plain as then the slope angle will be zero. Spatial variation of topographic factor for the months of June and December with lowest and highest solar zenith is represented in Fig. 4.

In the northern hemisphere, the topographic factor for a south facing terrain is more than 1.0 because the sunshine duration for south facing is longer and then again the topographic factor for north facing is less than 1.0 as ' n ' is shorter. Fig. 5 demonstrate the variation of TF with aspect in summer and winter season. Although in summer aspect does not seem to have a substantial influence on incoming insolation, but in winter TF is far larger for surface aspect from 120° to 240° than the rest. Same as aspect, the effect of slope is also far more significant in winter than summer. In north facing, the topographic factor tended to be inversely proportional to slope, while in case of south, it is totally the opposite i.e. TF is directly proportional to slope angle.

When plotted against the slope angle (Fig. 6) the quarterly mean TF also revealed its dependency on the sun's position in the sky. The rate of change in TF with slope angle for north facing surface is highest in winter (November–December–January) and lowest in summer (May–June–July) and is completely reverse for the south facing areas. The synergistic effect of slope and aspect on the topographic factor is given in Fig. 7.

3.3. Distribution of solar radiation

Low validation error as well as high correlation coefficient imply the statistical significance of the empirical models using sunshine durations and so are used to estimate the solar radiation at horizontal surface for 81 stations for the year of 2013. Due to South Korean complex topography, a topographical factor is applied before using the modeled data for interpolation. Special variability of mean monthly solar radiation is mapped using ordinary kriging method in a $30 \text{ m} \times 30 \text{ m}$ scale and given in the Fig. 8. The maps visibly indicate the areas receive high and low amount of radiation over South Korean peninsula. Daily mean solar radiation is found to be highest in May and lowest in December.

It is interesting to note that even in summer season, some areas of Korean peninsula receive very low amount of radiation due to high cloud conditions and geographical positions. Seasonal trend of solar insolation is shown in Figs. 9 and 10.

Due to the monsoon season starting with June and continuing for almost 30 days, the amount of cloud and precipitation is higher during these months. For these reason although the extraterrestrial radiation is higher in June in summer solstice, the observed radiation at ground level is higher during May with highest mean sunshine hours (7.78 h). The observed solar radiation data for 2013 shows a normal distribution with a mean

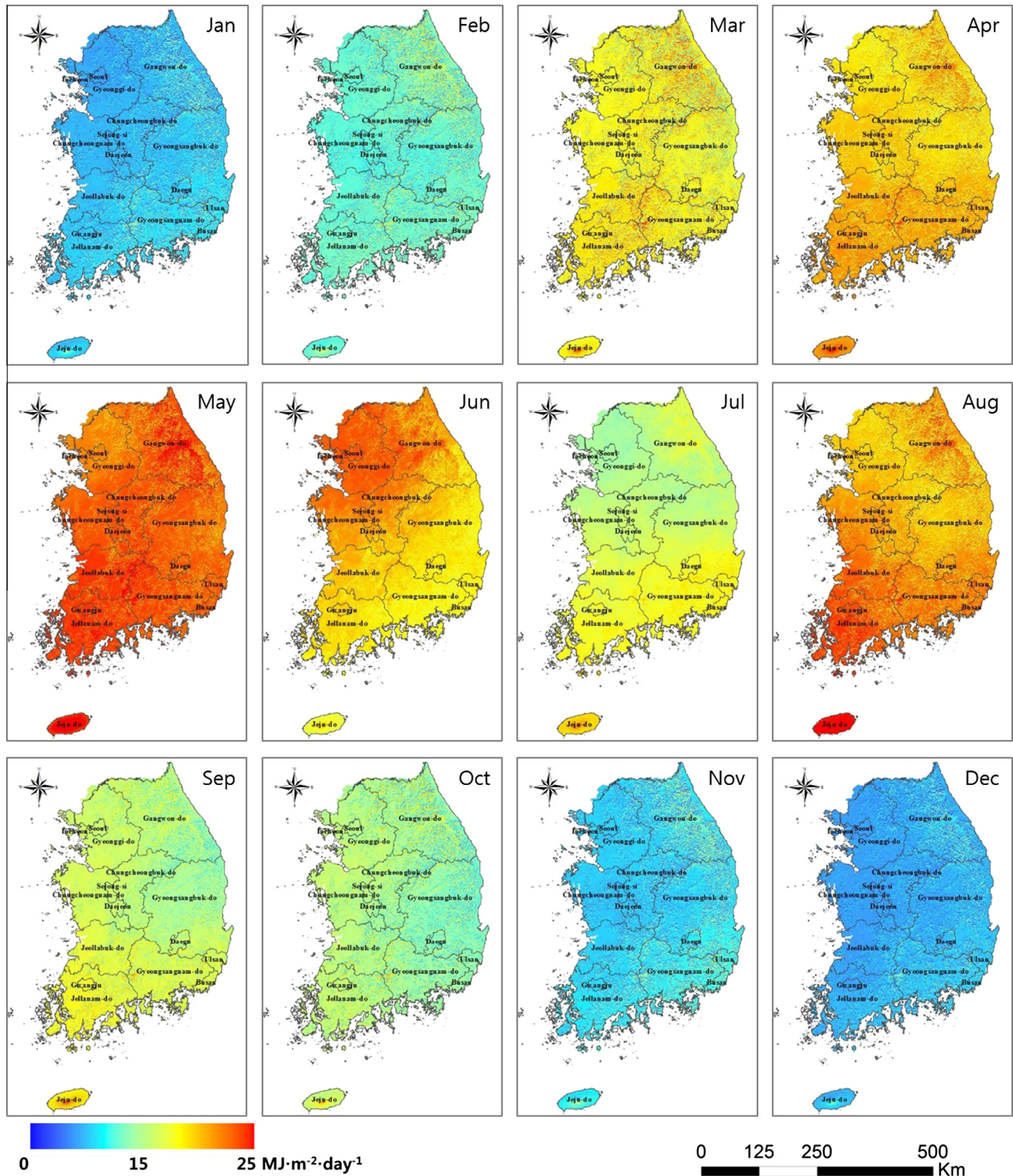


Fig. 8. Spatial distribution of monthly average daily incoming solar radiation considering surface slope and aspect in 2013.

$14.09 \pm 2.34 \text{ MJ m}^{-2} \text{ day}^{-1}$. Among which only 5% of observed value is higher than $25 \text{ MJ m}^{-2} \text{ day}^{-1}$ while more than 25% of the data is below $10 \text{ MJ m}^{-2} \text{ day}^{-1}$.

4. Conclusions

Now a days there is an increasing demand of accurate solar radiation data not only for the simulation for photovoltaic system but also as an input variable in many climatic and biogeochemical

models. Despite of its unavailability in maximum areas, solar radiation can be estimated from most commonly measured meteorological variables like sunshine duration, temperature etc. In South Korea 22 stations measured daily solar radiation and another 59 stations measured bright sunshine hours. Low cross validation errors and high correlation between observed and modeled dataset indicates the robust performance of empirical models based on sunshine duration used in the present study. The scenario of topographical influence on incoming insolation is utterly different in

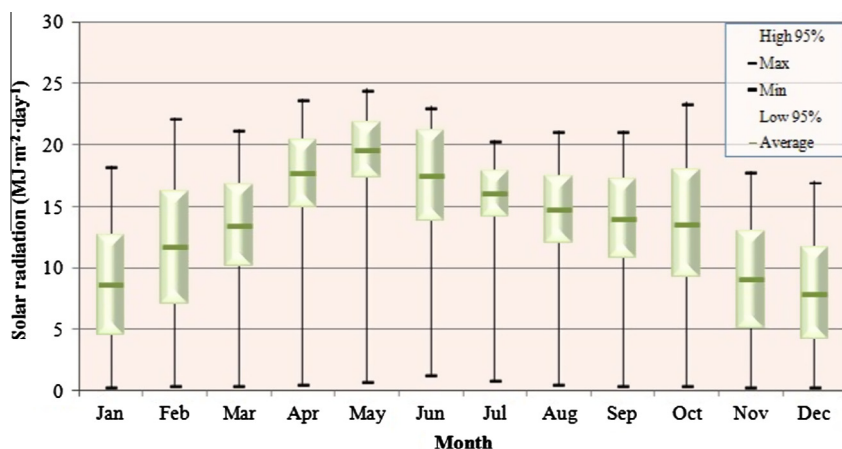


Fig. 9. Seasonal trend of monthly average daily incoming solar radiation in 2013.

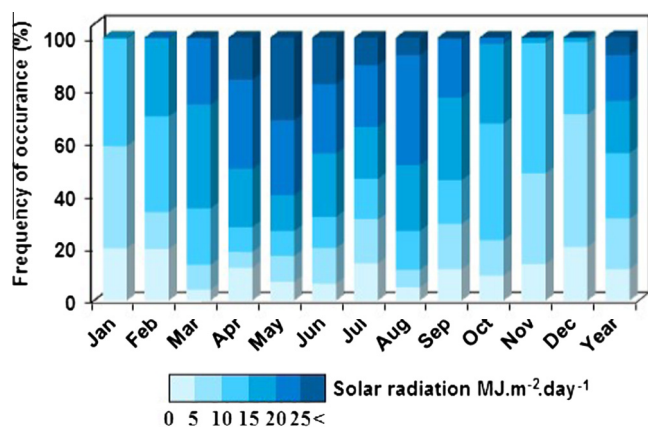


Fig. 10. Frequency distribution of observed solar radiation.

north and south facing terrains. In case of south facing areas, during winter TF is directly proportional to the surface slope angel which is entirely opposite for a north facing landscape. Both the effect of slope and aspect in summer is not as significant as in winter. Average global radiation and sunshine hours varies from $13.88 \text{ MJ m}^{-2} \text{ day}^{-1}$ and 5.13 h at southeast most station Gosan (Lat: 33.29°N) to $13.33 \text{ MJ m}^{-2} \text{ day}^{-1}$ and 5.45 h in northwest station Chuncheon (Lat: 37.90°N). Geographically South Korea is bounded by sea from three directions which gives it a mix oceanic and continental climatic characteristic with average precipitation and cloud coverage way more than the world average. This wet temperate climate associated with divert topography plays the main role in spatial variability of solar radiation overtrumping the latitudinal influence. Although the sun's zenith angle is lowest in June, due to monsoon season, the amount of incoming radiation is poorer in June than in May. For further study, other explanatory variable can be included in the empirical models and spatial interpolation which although will complicate the estimation method, may reduce the disagreement between observed and predicted values.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.enconman.2015.04.021>.

References

- [1] Technology Roadmap: Solar Photovoltaic Energy. International Energy Agency (IEA); 2010.
- [2] Zelenka A, Czeplak G, Agostino VD, Weine J, Maxwell E, Perez R. Techniques for supplementing solar radiation network data. In Report IEA Task 9. 2, Report No. IEA-SHCP-9D-1; 1992.
- [3] Santabarbara JM, Calbo J, Baldasano JM. Month-to-month variation of global solar radiation in Catalonia (Spain). *Int J Climat* 1996;16:711–21.
- [4] Mubiru J, Karume K, Majaliwa M, Banda EJK, Otiti T. Interpolating methods for solar radiation in Uganda. *Theor App Climat* 2007;88(3–4):259–63.
- [5] Oh SY, Park JK, Park JH. Evaluating the spatial and temporal solar energy potential in South Korea. *Proc Asia SPIE* 2012;85: 29–19.
- [6] Yaiche MR, Bouhanik A, Bekkouche SMA, Malek A, Benouaz T. Revised solar maps of Algeria based on sunshine duration. *Energy Convers Manage* 2014;82:114–23.
- [7] Wu GF, Liu YL, Wang TJ. Methods and strategy for modeling daily global solar radiation with measured meteorological data – a case study in Nanchang station, China. *Energy Convers Manage* 2007;48(9):2447–52.
- [8] Thornton PE, Running SW. An improved algorithm for estimating incident daily solar radiation from measurements of temperature, humidity, and precipitation. *Agric Forest Meteorol* 1999;199(93):211–28.
- [9] Bristow KL, Campbell GS. On the relationship between incoming solar radiation and daily maximum and minimum temperature. *Agric Forest Meteorol* 1984;31:159–66.
- [10] Meza F, Varas E. Estimation of mean monthly solar global radiation as a function of temperature. *Agric Forest Meteorol* 2000;100:231–41.
- [11] Weiss A, Hays CJ. Simulation of daily solar irradiance. *Agric Forest Meteorol* 2004;123:187–99.
- [12] Martí P, Gasque M. Improvement of temperature-based ANN models for solar radiation estimation through exogenous data assistance. *Energy Convers Manage* 2011;52(2):990–1003.
- [13] Yacef R, Mellit A, Belaid S, Şen Z. New combined models for estimating daily global solar radiation from measured air temperature in semi-arid climates: application in Ghardaia, Algeria. *Energy Convers Manage* 2014;2014(79):606–15.
- [14] Nielsen L, Prahm L, Berkowicz R, Conradsen K. Net incoming radiation estimated from hourly global radiation and/or cloud observations. *J Climat* 1981;1:255–72.
- [15] Supit I, van Kappel RR. A simple method to estimate global radiation. *Solar Energy* 1998;63:147–60.
- [16] Ehnberg JSG, Bollen MHJ. Simulation of global solar radiation based on cloud observations. *Solar Energy* 2005;78:157–62.
- [17] Angstrom A. Solar and terrestrial radiation. *Quart J Royal Meteorol Soc* 1924;50:121–5.
- [18] Prescott JA. Evaporation from a water surface in relation to solar radiation. *Trans Royal Soc South Australia* 1940;64:114–25.
- [19] Akpabio LE, Etuk SE. Relationship between global solar radiation and sunshine duration for one, Nigeria. *Turk J Phys* 2003;2003(27):161–7.
- [20] Wang Y, Zhang L. Relationship between global solar radiation and sunshine duration for Northwest China. *Int J Phys Sci* 2010;5(7):1023–33.

- [21] Jo DK, Yun CY, Kim KD, Kang YH. A study on the estimating solar radiation using hours of bright sunshine for the installation of photovoltaic system in Korea. *J Korean Solar Energy Soc* 2011;31(4):72–9.
- [22] Liu J, Linderholm HW, Chen D, Yu Q, Wu D, Haginoya S. Observation and calculation of the solar radiation on the Tibetan Plateau. *Energy Convers Manage* 2012;57:23–32.
- [23] Park JK, Oh SY, Back SC, Kim BS, Shin HS, Park JH. Estimation of monthly solar radiation using sunrise duration at Daegwallyeong. *J Agric Sci Res* 2013;29(1):28–32.
- [24] Li MF, Liu HB, Guo PT, Wu W. Estimation of daily solar radiation from routinely observed meteorological data in Chongqing, China. *Energy Convers Manage* 2010;51(2):2575–9.
- [25] Iziomon MG, Mayer H. Assessment of some global solar radiation parameterizations. *J Atmos Solar-Terres Phys* 2002;2002(64):1631–43.
- [26] Trnka M, Zalud Z, Eitzinger J, Dubrovsky M. Global solar radiation in Central European lowlands estimated by various empirical formulae. *Agric Forest Meteorol* 2005;131:54–76.
- [27] Al-Mostafa ZA, Maghrabi AH, Al-Shehri SM. Sunshine-based global radiation models: a review and case study. *Energy Convers Manage* 2014;84:209–16.
- [28] Qiu XF, Zeng Y, Liu SM. Distributed modeling of extraterrestrial solar radiation over rugged terrain. *Chin J Geophys* 2005;48:1100–7.
- [29] Kumar L, Skidmore AK, Knowles E. Modelling topographic variation in solar radiation in a GIS environment. *Int J Geographic Inform Sci* 1997;11:475–97.
- [30] Pons X, Miquel N. Mapping a topographic global solar radiation model implemented in a GIS and refined with ground data. *Int J Climat* 2008;28:1821–34.
- [31] Ruiz-Arias JA, Tovar-Pescador J, Pozo-Vázquez D, Alsamamra H. A comparative analysis of DEM-based models to estimate the solar radiation in mountainous terrain. *Int J Geographic Inform Sci* 2009;23:1049–76.
- [32] Yun JI. A simple method using a topography correction coefficient for estimating daily distribution of solar irradiance in complex terrain. *Korean J Agric Forest Meteor* 2009;11(1):13–8.
- [33] Akinoglu BG, Ecevit A. Construction of a quadratic model using modified Angstrom coefficients to estimate global solar radiation. *Solar Energy* 1990;45:85–92.
- [34] Duffie JA, Beckman WA. *Solar Engineering of Thermal Processes*. New York, U.S.A.: John Wiley & Sons; 1991.
- [35] Korean Meteorological Administration. *Annual Report*; 2011.
- [36] Ismail MS, Moghavvemi M, Mahlia TMI. Analysis and evaluation of various aspects of solar radiation in the Palestinian territories. *Energy Convers Manage* 2013;73:57–68.
- [37] Hottel HC, Woertz BB. Evaluation of flat solar heat collector. *ASME Trans* 1942;64:91.
- [38] Liu BYH, Jordan RC. The interrelationship and characteristic distribution of direct, diffuse and total solar radiation. *Solar Energy* 1960;4(3):1–19.
- [39] Merino GG, Jones D, Stooksbury DE, Hubbard KG. Determination of semivariogram models to Krige hourly and daily solar irradiance in Western Nebraska. *J Appl Meteorol* 2001;40:1085–94.